Activity Diagram

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01
Introduction to Activity Diagram

Overview of Activity Diagram



What is an Activity Diagram?

A UML diagram describing dynamic aspects of a system.
Represents **flow of control from one operation to another.**
Flow can be sequential, branched, or concurrent.
Uses elements like forks and joins to manage flow control.

Benefits of Activity Diagrams

Key Benefits

- Demonstrates algorithm logic.
- Describes steps in UML use cases.
- Illustrates business processes or workflows between **users and systems.**
- Simplifies and clarifies **complex use cases.**
- Helps model software architecture elements (methods, functions, operations).



02
Basic Components of an Activity Diagram

Start and End Nodes

Start Node

Represents the beginning of an activity.
Symbolized by a solid black circle.



Start symbol

End Node

Represents the final step in the activity.
Shown as an outlined black circle.



End symbol

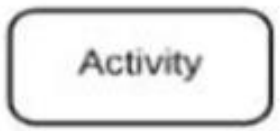
Actions and Activity Symbols

Action

A step performed by users or software.
Represented by round-edged rectangles.

Activity Symbol

Illustrates activities that constitute the modeled process.
Includes short descriptions inside the shape.



Activity symbol

Control Flows and Connectors

Control Flow

- Connectors showing directional flow between steps.

- Incoming arrow starts a step; outgoing arrow continues flow after completion.



Connector symbol

Decision Nodes

Decision Node

- Conditional branch represented by a diamond shape.
- Has one input and multiple outputs.
- Each output path includes condition text explaining branching options.

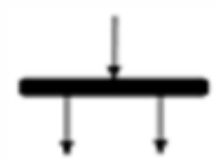


Decision symbol

Fork and Join Symbols

Fork Symbol

- Splits one activity flow into multiple concurrent flows.
- Symbolized by multiple arrowed lines emerging from a thick line.



Fork symbol

Join Symbol

- Combines multiple concurrent activities into one sequential flow.
- Represented by a thick vertical or horizontal line.



Joint symbol/ Synchronization bar



03
Examples

Example: Activity diagram for a login page

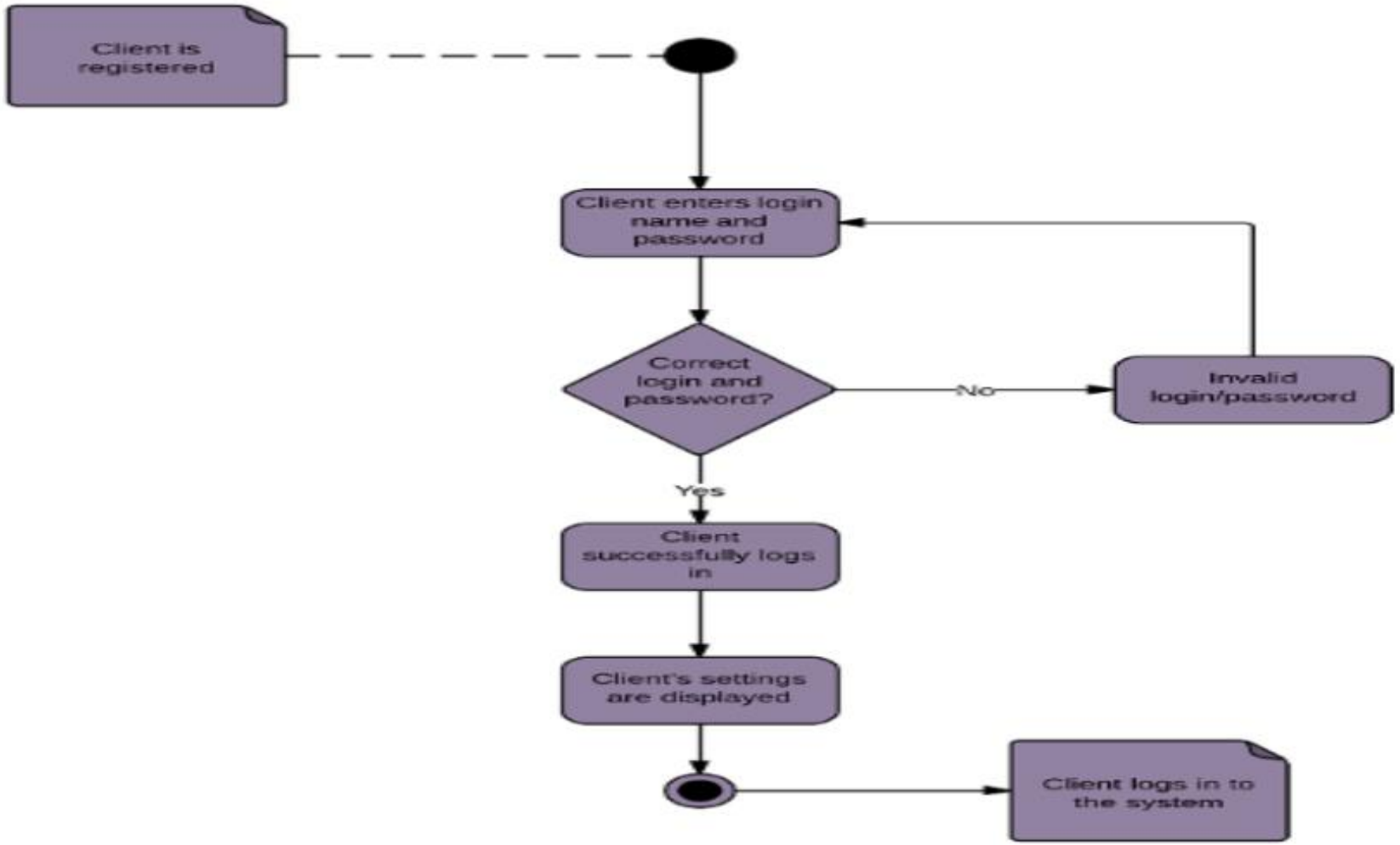
Many of the activities people want to accomplish online—such as:

- checking email,
- managing finances,
- ordering clothes, etc.—
- require them to **log into a website**.

This **activity diagram** shows the process of logging into a website, from:

- entering a username and password
- to successfully logging in to the system.

Example: Activity diagram for a login page



Example: Activity diagram for a banking system

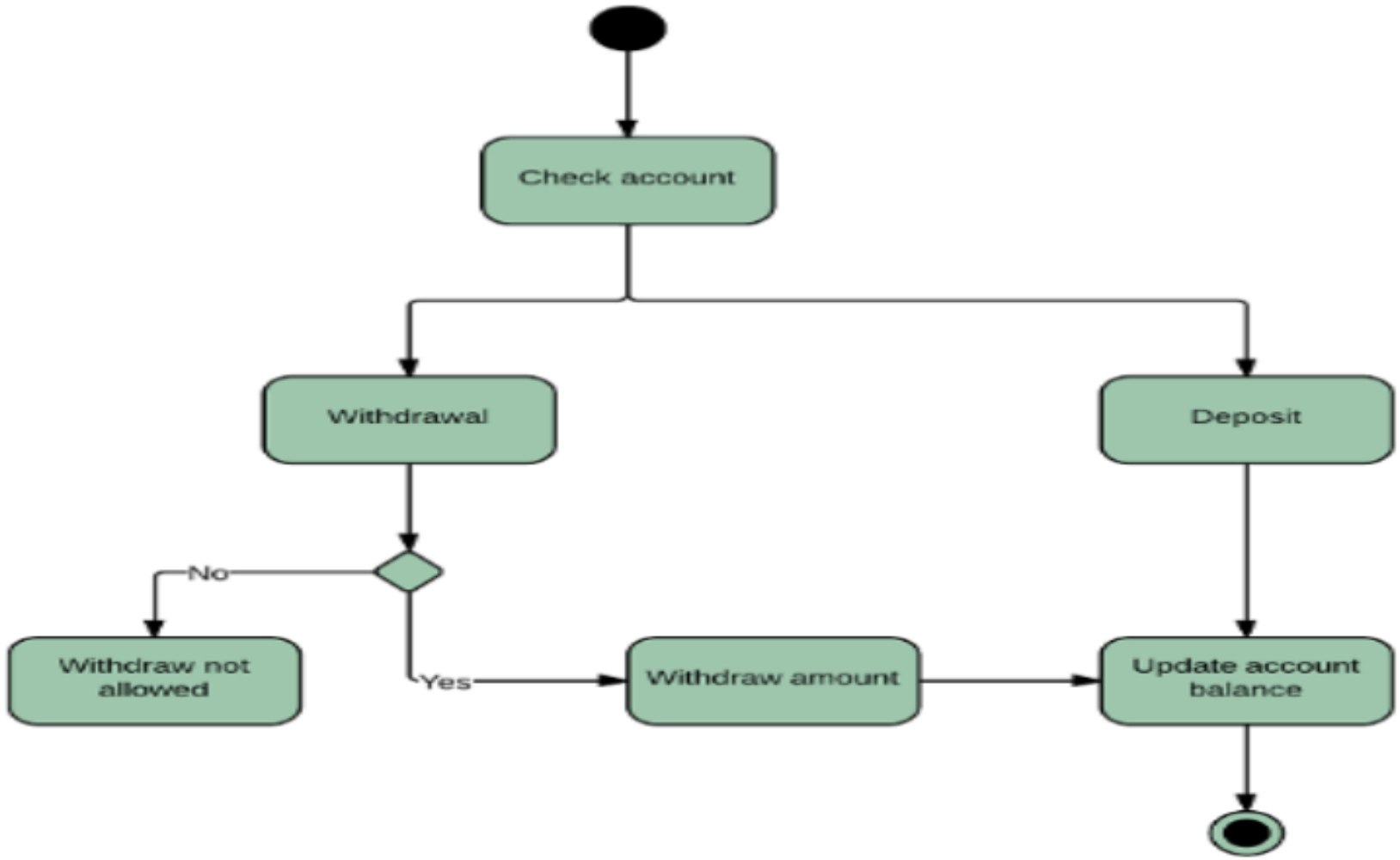
This **diagram** shows the process of:

- withdrawing money from, or
- depositing money into
- a **bank account**.

An advantage of representing the **workflow visually in UML** is the ability to:

- show **withdrawals** and **deposits** together on one chart.

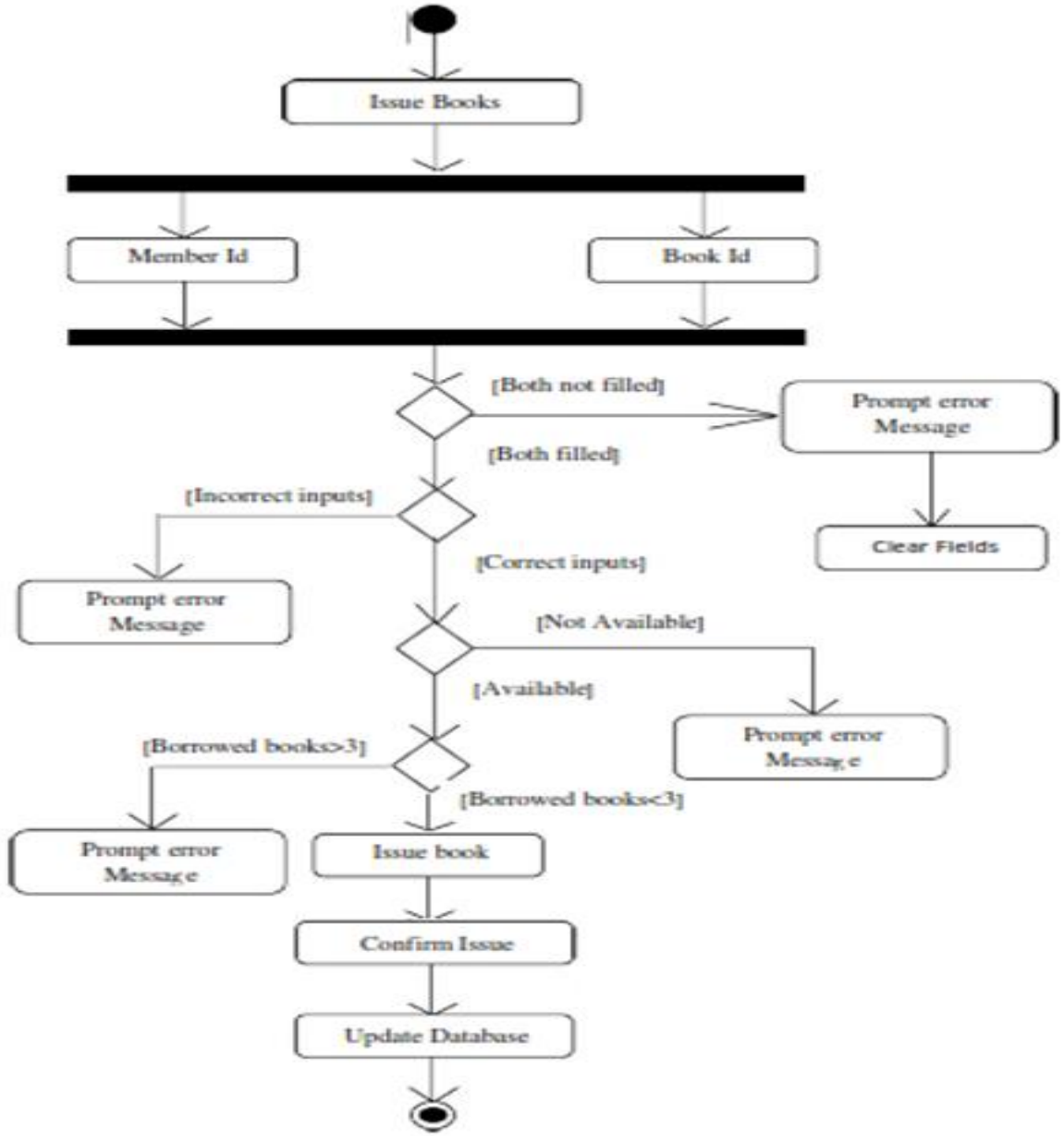
Example: Activity diagram for a banking system



Example: Activity diagram for the "Issue book" use case

1. **Start**
2. **Enter Member ID and Book ID**
3. **Check if both fields are filled**
    1. If not → Display “Enter all fields” message → **End**
4. **Validate Member ID and Book ID**
    1. If invalid → Display “Invalid Member or Book ID” → **End**
5. **Check Book Availability**
    1. If not available → Display “Book Not Available” → **End**
6. **Check Number of Books Issued by Member**
    1. If 3 or more → Display “Limit Exceeded” → **End**
7. **Issue Book**
    1. Record transaction and assign book to member
8. **Update Modification**
    1. Update database (book status, member record)
9. **Display Success Message**
10. **End**

Example: Activity diagram for the "Issue book" use case



Task – Activity Diagram for Student Registration System

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Scenario:

Students register for courses each semester using a university system.
You have already drawn the **Use Case Diagram** showing how **students** and **admins** interact with the system.

Task Description:

Create an **Activity Diagram** that represents the **“Register Course”** process in the Student Registration System.

Thanks

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